

Telco Customer Churn Analysis

An End-to-End Data Analytics Project

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Tools Used: Python (Pandas, Matplotlib, Seaborn) | Power BI | Google Colab

1. Problem Statement

1.1 Background

A telecom company serves thousands of customers across monthly and yearly subscription plans. Customer acquisition is expensive, making retention of existing customers far more cost-effective than acquiring new ones.

1.2 Problem

The company is experiencing significant customer churn, with a large number of customers cancelling their subscriptions. Management does not have visibility into which customers are leaving, why they are leaving, or which factors are driving this churn.

1.3 Objective

To analyze customer data in order to identify patterns and key drivers of churn, understand which customer segments are most at risk, and provide data-backed recommendations to reduce the overall churn rate.

1.4 Success Criteria

- A dashboard showing the overall churn rate and churn breakdown by key segments
- At least three specific, data-backed insights explaining why customers churn
- At least three actionable, numbers-driven business recommendations

1.5 Business Questions Addressed

- What is the overall churn rate of the company?
- Which contract type has the highest churn — monthly, one-year, or two-year?
- Do customers who have been with the company longer churn less?
- Which services are associated with higher churn — internet, phone, streaming?
- Does payment method affect whether a customer churns?
- What is the revenue impact of customers who churned?
- Which factors most strongly predict customer churn?
- Which customer segments are at the highest risk of churning?
- What retention strategies can reduce churn in high-risk segments?

2. Dataset Overview

The analysis uses the Telco Customer Churn dataset, sourced from Kaggle. It contains 7,043 customer records across 21 attributes covering demographics, account information, subscribed services, billing details, and churn status.

Attribute	Detail	Notes
Source	Kaggle - Telco Customer Churn	Public dataset
Rows	7,043 customers	No duplicates found

Columns	21 (+1 engineered)	tenure_group added
Target Variable	Churn (Yes/No)	26.54% churn rate

3. Data Cleaning & Preparation

Before analysis, the raw dataset was checked for data quality issues using Python (Pandas). The following issues were identified and resolved:

Issue Found	Action Taken	Reason
TotalCharges stored as text with 11 blank entries	Converted to numeric (float); blanks replaced with 0	All 11 affected customers had tenure = 0 (newly joined, not yet billed)
SeniorCitizen stored as 0/1	Converted to Yes/No	Consistency with other categorical columns
No tenure segmentation	Created new column 'tenure_group' (0-12, 12-24, 24-48, 48+ months)	Enables segment-level churn analysis
Duplicate rows	Checked - none found	Data integrity confirmed
Categorical value consistency	Checked all unique values across 17 categorical columns - no typos or inconsistencies found	Ensures accurate grouping in analysis

4. Exploratory Data Analysis - Key Findings

Each business question was answered using grouped aggregations and visualizations in Python (Matplotlib and Seaborn). The key findings are summarized below.

4.1 Overall Churn Rate

26.54% of customers have churned - 1,869 out of 7,043 total customers. The industry average for telecom churn is typically 15-20%, meaning this company's churn rate is significantly above benchmark and represents a serious retention concern.

4.2 Churn by Contract Type

Contract type is the strongest predictor of churn in the dataset:

- Month-to-month contracts: 42.71% churn rate
- One-year contracts: 11.27% churn rate
- Two-year contracts: 2.83% churn rate

Month-to-month customers churn at nearly 15 times the rate of two-year contract customers, indicating that lack of long-term commitment is strongly associated with customer attrition.

4.3 Churn by Tenure

Churn rate decreases consistently as customer tenure increases:

- 0-12 months: 47.44% churn (critical zone)
- 12-24 months: 28.71% churn
- 24-48 months: 20.39% churn
- 48+ months: 9.51% churn (loyal zone)

Nearly 1 in 2 new customers leaves within their first year, identifying the first 12 months as the most critical period for retention efforts.

4.4 Churn by Service Type

Internet service type and add-on services show a strong relationship with churn:

- Fiber optic internet customers: 41.89% churn (highest of all services)
- DSL customers: 18.96% churn
- Customers with no internet service: 7.40% churn (lowest)
- Customers without Online Security: 41.77% churn vs 14.61% for those with it
- Customers without Tech Support: 41.64% churn vs 15.17% for those with it

Fiber optic customers, who typically pay premium prices, churn at the highest rate - especially when they lack protective add-ons such as security and tech support, suggesting dissatisfaction with perceived value.

4.5 Churn by Payment Method

- Electronic check: 45.29% churn (highest)
- Mailed check: 19.11% churn

- Bank transfer (automatic): 16.71% churn
- Credit card (automatic): 15.24% churn (lowest)

Customers on automatic payment methods churn at roughly one-third the rate of electronic check users. Automatic payments appear to create passive commitment, while manual monthly payments give customers a recurring opportunity to reconsider their subscription.

4.6 Revenue Impact of Churn

- Total revenue lost to churned customers: \$2,862,926.90 (17.83% of total revenue)
- Monthly recurring revenue at risk: \$139,130.85
- Average monthly charge - churned customers: \$74.44
- Average monthly charge - retained customers: \$61.27

Churned customers paid more per month on average than retained customers, meaning the company is disproportionately losing its highest-value customers. A 10% reduction in churn would save approximately \$286,000 in revenue.

5. Power BI Dashboard

An interactive Power BI dashboard was built to present the churn analysis to business stakeholders. It includes 5 KPI cards summarizing the overall situation, four charts breaking down churn by contract, tenure, internet service and overall distribution, and a slicer enabling filtering by contract type.

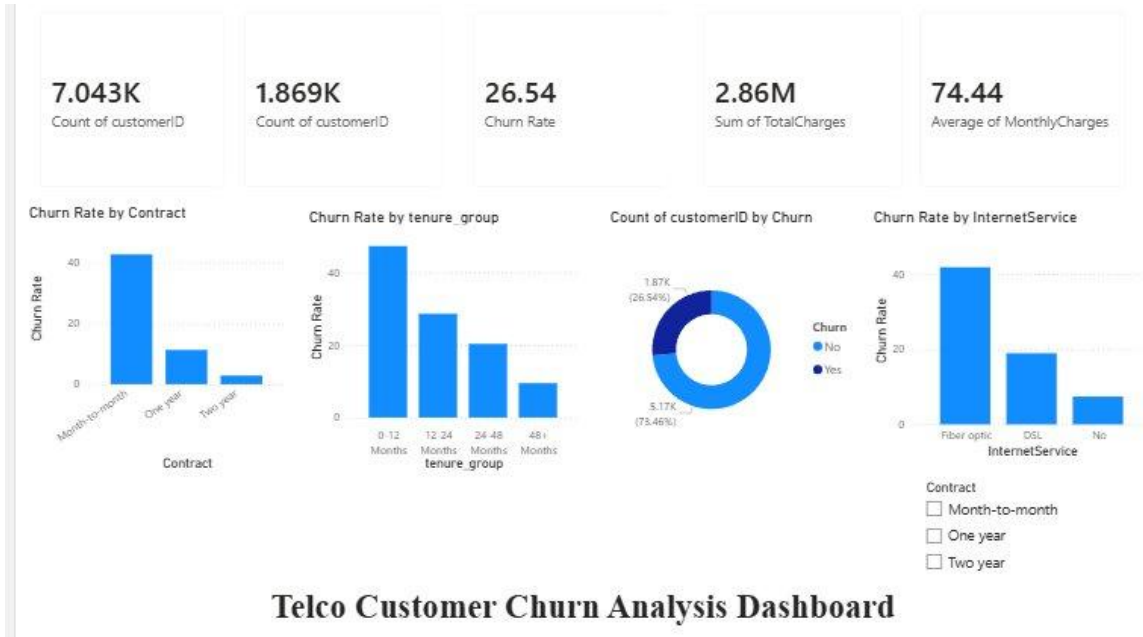


Figure 1: Telco Customer Churn Analysis Dashboard (Power BI)

6. Recommendations

Based on the findings above, the following data-backed recommendations are proposed to reduce customer churn:

6.1 Convert Month-to-Month Customers to Longer Contracts

Month-to-month customers churn at 42.71% versus 2.83% for two-year contracts. Offering a discount or loyalty incentive for switching to a one or two-year contract could significantly reduce churn among this segment, which represents the largest share of churned customers.

6.2 Focus Retention Efforts on the First 12 Months

Nearly half of new customers (47.44%) churn within their first year. Introducing a structured onboarding program, proactive check-ins, or a loyalty discount at the 3-month mark could help retain customers through this high-risk period.

6.3 Bundle Protective Add-ons with Fiber Optic Plans

Fiber optic customers without Online Security or Tech Support churn at over 41%, compared to ~15% for those with these add-ons. Including these services as part of the standard fiber optic package - rather than as paid extras - could improve perceived value and reduce churn.

6.4 Encourage Migration to Automatic Payment Methods

Electronic check users churn at 45.29%, roughly three times the rate of automatic payment users. A small incentive (e.g., a one-time bill credit) for switching to automatic bank transfer or credit card payments could reduce the monthly 'reconsideration' moment that appears linked to higher churn.

6.5 Prioritize High-Risk Segments

The combination of month-to-month contract, tenure under 12 months, and fiber optic internet without add-ons represents the company's highest-risk customer profile. This segment should be the primary target for all retention campaigns, as it offers the highest potential return on retention investment.

7. Conclusion

This analysis identified contract type, customer tenure, internet service type, and payment method as the strongest drivers of customer churn for this telecom company. With an overall churn rate of 26.54% and \$2.86 million in lost revenue, targeted retention strategies focused on month-to-month customers in their first year - particularly fiber optic subscribers without protective add-ons - represent the highest-impact opportunity to improve customer retention and protect recurring revenue.

7.1 Tools & Techniques Used

- Python (Pandas, NumPy) - data cleaning and feature engineering
- Matplotlib, Seaborn - exploratory data analysis and visualization
- Power BI - interactive dashboard development with DAX measures

- Google Colab - cloud-based notebook environment
- GitHub - version control and project documentation